

# Path Algebras of Quivers as a Sparse Multi-Hop Attention for Mitigating Over-Squashing in GNNs

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**Abstract.** Message-passing graph neural networks suffer from *over-squashing*: signal travelling along many long paths into a node is compressed into one fixed-width vector and lost. We introduce **Walk Attention**: the *path algebra*  $\mathbb{k}Q$  of the underlying quiver supplies, through its graded components, the exact multi-hop reachability structure, which we use as the sparse *support* of a learned attention—the algebra says which vertices may interact at range  $k$ , the network learns how much. On bottleneck graphs with tunable over-squashing we *prove* the separation—message passing with linear aggregation is at exactly chance accuracy for every depth and width, while a single Walk Attention layer at the matching range solves the retrieval task exactly—and confirm it empirically ( $1.000 \pm 0.000$  over five seeds, where one-hop attention degrades toward chance); Walk Attention also stays exact on RINGTRANSFER and, under a shared small budget, ties one-hop attention on Peptides-func; an algebraic *severity index*—the excess of incoming walk entropy over representation width—predicts this regime dependence (advantage exactly zero below a threshold, large above it, over eleven configurations) and selects a real citation network on which Walk Attention is the best of the compared models. A negative result sharpens the design: collapsing equivalent walks via the quotient  $\mathbb{k}Q/I$  discards multiplicity the task needs; the algebra’s role is to *define the support*, not to compress. Walk Attention is the learned reading of an algebraic substrate whose dynamical reading is the automata-of-impact framework.

**Keywords:** Graph neural networks · Over-squashing · Path algebra · Quiver · Representation theory · Attention.

## 1 Introduction

Graph neural networks built on the message-passing paradigm [11] update each node by aggregating messages from its immediate neighbours and iterating this local rule across layers, so that after  $k$  layers information from nodes  $k$  hops

away reaches a target node. While simple and effective, the scheme has a well-documented failure mode: *over-squashing* [3]. When the number of paths carrying information into a node grows—typically exponentially with distance—the messages along all of those paths must be compressed into a single fixed-width vector; the receiving node cannot recover them individually, and long-range signal is lost. Over-squashing is the principal obstacle to learning tasks whose labels depend on distant or globally-distributed information. Existing remedies act on the graph or on the aggregation: *rewiring* widens bottlenecks [19], curvature and sensitivity analyses explain *where* squashing occurs [7], and a complementary line replaces local aggregation by global Transformer attention [20], at the cost of an all-pairs interaction that ignores graph structure.

In this paper we take a different route, grounded in the representation theory of quivers. A *quiver* is a directed graph, possibly with multiple arrows [17,4], and its *path algebra*  $\mathbb{k}Q$  has the directed walks of the quiver as a basis, with concatenation as multiplication. The graded piece  $e_j \cdot \mathbb{k}Q_k \cdot e_i$  of this algebra has a basis consisting of the directed walks of length  $k$  from vertex  $i$  to vertex  $j$ ; its dimension equals  $(A^k)_{ij}$ , the corresponding entry of the  $k$ -th power of the adjacency matrix. The path algebra therefore provides an *exact, algebraic description of multi-hop reachability and multiplicity*.

Our central observation is that this structure is precisely what is needed to define a principled, sparse, multi-hop attention. We let each node attend directly to every vertex reachable from it by a length- $k$  walk: the algebra supplies the *support* of the attention (which pairs may interact at range  $k$ ), and the network learns the *weights* (how much). Because the walk-reachability support is sparse—typically a small fraction of all node pairs—this is far cheaper than full self-attention, while still acting at a distance and thereby bypassing the iterative bottleneck that causes over-squashing. We call the resulting architecture **Walk Attention**. Since a graph network on  $Q$  is a representation of the quiver ( $\text{Rep}(Q) \cong \mathbb{k}Q\text{-Mod}$ ), Walk Attention is the *learned* reading of an algebraic substrate whose *dynamical* reading is the automata-of-impact-over-quivers framework [25] (Section 3.6).

### *Contributions.*

1. A self-contained account (Section 3) of quivers, path algebras, the impact-degree neighbourhood system and the *walk operators* encoding multi-hop reachability.
2. **Walk Attention** (Section 4): a learned multi-hop attention whose support is the path-algebra walk-reachability structure, related to message passing and to Transformer attention.
3. A *proved* separation (Section 4.2): with the standard one-layer-per-hop depth, any GNN whose first aggregation is linear in the source features (GCN, GIN) has exactly chance accuracy on the bottleneck family, for every depth, width and parameters, while a single Walk Attention layer at the matching range solves the task exactly—confirmed empirically (Section 5), together with a *severity index*  $\Psi$ , computed from the graded dimensions, that predicts the

regime-dependent advantage on RINGTRANSFER and Peptides-func and selects a real citation network with severe squashing.

4. A *negative* result: collapsing equivalent walks via the quotient  $\mathbb{k}Q/I$  discards needed multiplicity and fails; the quotient’s role is to shape the support, not to compress.

## 2 Related Work

*Over-squashing, attention, and multi-hop aggregation.* Alon and Yahav [3] identified over-squashing with the NEIGHBORMATCH probe; Topping et al. [19] connected it to negative curvature and proposed SDRF rewiring; Di Giovanni et al. [7] related it to width, depth and topology. GAT [21] learns one-hop attention weights; graph Transformers apply all-pairs attention [20] at  $O(n^2)$  with positional encodings. Multi-hop aggregation is well explored: MixHop [2] and SIGN [9] mix adjacency powers, diffusion convolution [10] and MAGNA [22] diffuse weights or attention scores, shortest-path message passing [1], NAGphormer [6] and Graphormer [23] aggregate or bias by distance, and DRew [12] rewires with delay. Walk Attention differs in the object defining the support: the *graded path algebra*—walks with their multiplicity, not distance shells or diffusion weights—which makes the support exact (Proposition 1), exposes the quotient  $\mathbb{k}Q/I$  as a design degree of freedom, and embeds the architecture in quiver representation theory.

*Path-algebraic structure.* The path algebra is a standard object in representation theory [17,4]; path-algebraic constructions have recently been applied to combinatorial problems [16,15]. Closest to this work is the impact-degree neighbourhood system and the automata-of-impact framework [25,13], the dynamical sibling of Walk Attention over the same substrate (Section 3.6). What is new, to our knowledge, is taking the support from the graded path algebra, with its multiplicities, its quotient, and its dynamical reading.

## 3 The Path Algebra of a Quiver

This section builds, assuming no prior exposure, the quivers, paths, graded path algebra, impact-degree neighbourhood system and walk operators the model relies on; standard references are [17,4].

### 3.1 Quivers and paths

**Definition 1.** A quiver is a quadruple  $Q = (Q_0, Q_1, s, t)$  where  $Q_0$  is a finite set of vertices,  $Q_1$  a finite set of arrows, and  $s, t: Q_1 \rightarrow Q_0$  assign to each arrow  $\alpha$  its source  $s(\alpha)$  and target  $t(\alpha)$ ; we write  $\alpha: i \rightarrow j$ . Multiple arrows between the same ordered pair are allowed. We identify  $Q_0$  with  $\{1, \dots, n\}$ .

A quiver is thus a directed multigraph: entities, directed relations, and parallel arrows recording multiplicity.

*Example 1.* Let  $Q_0 = \{1, 2, 3\}$  with arrows  $\alpha, \alpha': 1 \rightarrow 2$  (two parallel arrows),  $\beta: 2 \rightarrow 3$ , and a loop  $\gamma: 3 \rightarrow 3$ . Its adjacency matrix, counting arrows, is

$$A = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

In a citation network the vertices are documents, arrows are citations,  $\alpha, \alpha'$  a double citation and  $\gamma$  a self-citation.

**Definition 2.** A path of length  $k \geq 1$  in  $Q$  is a sequence of arrows  $p = \alpha_k \cdots \alpha_2 \alpha_1$  with  $t(\alpha_r) = s(\alpha_{r+1})$ , read right-to-left as in function composition, with source  $s(p) = s(\alpha_1)$  and target  $t(p) = t(\alpha_k)$ . Each vertex  $i$  also carries a trivial path  $e_i$  of length 0.

A path is a directed *walk*: vertices and arrows may repeat. In Example 1,  $\beta\alpha$  and  $\beta\alpha'$  are distinct paths of length 2 from 1 to 3, and the loop generates walks  $\gamma^m\beta\alpha$ . The number of walks of a given length between two vertices is the quantity that drives over-squashing.

### 3.2 The path algebra and its grading

Let  $\mathbb{k}$  be a field (in practice  $\mathbb{k} = \mathbb{R}$ ).

**Definition 3.** The path algebra  $\mathbb{k}Q$  is the  $\mathbb{k}$ -vector space with basis the set of all paths of  $Q$ , endowed with the bilinear multiplication given on basis paths by concatenation:

$$q \cdot p = \begin{cases} qp & \text{if } s(q) = t(p), \\ 0 & \text{otherwise.} \end{cases}$$

The trivial paths satisfy  $e_{t(p)}p = p = pe_{s(p)}$  and  $e_i e_j = \delta_{ij} e_i$ , so  $\{e_i\}$  is a complete set of orthogonal idempotents and  $1 = \sum_i e_i$  is the unit.

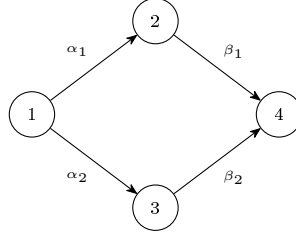
The multiplication respects length: concatenating paths of lengths  $k$  and  $\ell$  yields a path of length  $k + \ell$  (or zero). Hence  $\mathbb{k}Q$  is a *graded* algebra,

$$\mathbb{k}Q = \bigoplus_{k \geq 0} \mathbb{k}Q_k, \quad \mathbb{k}Q_k \cdot \mathbb{k}Q_\ell \subseteq \mathbb{k}Q_{k+\ell},$$

where  $\mathbb{k}Q_k$  is spanned by the paths of length exactly  $k$ ; in particular  $\mathbb{k}Q_0 = \bigoplus_i \mathbb{k}e_i$  and  $\mathbb{k}Q_1$  is spanned by the arrows. The idempotents let us isolate the paths between a fixed pair of vertices: the subspace  $e_j \cdot \mathbb{k}Q_k \cdot e_i$  is spanned exactly by the length- $k$  paths from  $i$  to  $j$ , and its dimension is therefore the number of such walks. The following proposition states the correspondence with linear algebra precisely; it underlies everything the model computes.

**Proposition 1.** Let  $A \in \mathbb{N}^{n \times n}$  be the adjacency matrix of  $Q$ ,  $A_{ij} = \#\{\alpha \in Q_1 : \alpha: i \rightarrow j\}$ . Then for all  $k \geq 0$  and all  $i, j \in Q_0$ ,

$$\dim_{\mathbb{k}}(e_j \cdot \mathbb{k}Q_k \cdot e_i) = (A^k)_{ij}.$$



**Fig. 1.** The diamond quiver of Example 2: two parallel length-2 walks  $1 \rightsquigarrow 4$ ;  $\dim(e_4 \mathbb{k} Q_2 e_1) = 2$ .

*Proof.* By induction on  $k$ ; for  $k \in \{0, 1\}$  this is the definition. Every length- $k$  path from  $i$  to  $j$  factors uniquely as  $\alpha \cdot p$ , where  $p$  has length  $k - 1$  from  $i$  to some intermediate vertex  $r$  and  $\alpha: r \rightarrow j$  is an arrow; the number of such factorisations is  $\sum_r (A^{k-1})_{ir} A_{rj} = (A^k)_{ij}$ .  $\square$

We call  $n_k(i, j) := (A^k)_{ij} = \dim_{\mathbb{k}}(e_j \mathbb{k} Q_k e_i)$  the *walk multiplicity* from  $i$  to  $j$  at range  $k$ . It is exactly the number of length- $k$  messages that a message-passing GNN forces from node  $i$  into node  $j$  after  $k$  layers—and hence the source of over-squashing.

*Example 2.* Let  $Q$  have vertices  $\{1, 2, 3, 4\}$  and arrows  $\alpha_1: 1 \rightarrow 2$ ,  $\alpha_2: 1 \rightarrow 3$ ,  $\beta_1: 2 \rightarrow 4$ ,  $\beta_2: 3 \rightarrow 4$  (Figure 1). Then

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

and  $\dim(e_4 \mathbb{k} Q_2 e_1) = (A^2)_{14} = 2$ , in agreement with Proposition 1: the two *parallel* length-2 walks  $\beta_1 \alpha_1$  and  $\beta_2 \alpha_2$ , with the same source, target and length. Whether they should be treated as one channel or two is a modelling decision we return to in Section 5.

### 3.3 Impact degree and the fundamental neighbourhood system

The grading induces a stratification of the vertices around each node. The *impact degree*  $g(i, j)$  is the length of the shortest directed path from  $i$  to  $j$  ( $g(i, i) = 0$ ;  $g(i, j) = \infty$  if none exists); equivalently  $g(i, j) = \min\{k : (A^k)_{ij} > 0\}$ .

**Definition 4.** For a node  $c \in Q_0$ , its in-neighbourhood of radius  $k$  is  $A_k(c) = \{i \in Q_0 : g(i, c) \leq k\}$ , the set of nodes that can influence  $c$  along a directed path of length at most  $k$ . The nested family  $\{A_k(c)\}_{k \geq 0}$ , with  $A_0(c) = \{c\} \subseteq A_1(c) \subseteq \dots$ , is the fundamental neighbourhood system (FNS) of  $c$ ; the layer  $\mathcal{N}_k(c) = A_k(c) \setminus A_{k-1}(c)$  collects the nodes at impact degree exactly  $k$ .

Its relation to walk-reachability is the following.

**Proposition 2.** *For  $k \geq 1$  let  $\mathcal{S}_k = \{(c, i) : (A^k)_{ic} > 0\}$  be the pairs joined by a length- $k$  walk  $i \rightsquigarrow c$ . Then*

$$\{(c, i) : i \in \mathcal{N}_k(c)\} \subseteq \mathcal{S}_k,$$

*and the inclusion is strict in general: length- $k$  walks may also reach nodes at impact degree  $< k$ . If  $Q$  is graded—there is  $\rho : Q_0 \rightarrow \mathbb{Z}$  with  $\rho(t(\alpha)) = \rho(s(\alpha)) + 1$  for every arrow  $\alpha$ —equality holds for every  $k$ .*

*Proof.* A shortest path of length  $k$  is a length- $k$  walk. Strictness: in  $1 \rightarrow 2 \rightarrow 3$  with an extra arrow  $1 \rightarrow 3$ ,  $(A^2)_{13} > 0$  yet  $g(1, 3) = 1$ . If  $Q$  is graded every walk  $i \rightsquigarrow c$  has length  $\rho(c) - \rho(i)$ .  $\square$

The bottleneck graphs of Section 5 are graded (the layer index is such a  $\rho$ ), so there the supports coincide with the FNS layers; on a ring  $\mathcal{S}_k$  strictly contains the layer relation. Stacking ranges  $k = 1, 2, \dots$  walks an embedding outward through the FNS, and  $n_k(i, c)$  records *how many* length- $k$  paths populate each layer.

### 3.4 Walk operators

Proposition 1 lets us turn the graded path algebra into a family of linear operators on node features, one per range; these are the bridge to the neural model.

**Definition 5.** *For each range  $k \geq 1$  define  $W^{(k)} \in \mathbb{R}^{n \times n}$  by*

$$W_{ji}^{(k)} = \frac{(A^k)_{ij}}{Z_k}, \quad Z_k = \max_j \sum_i (A^k)_{ij},$$

*a single normalising constant per range. The walk-reachability support at range  $k$  is  $\mathcal{S}_k = \{(j, i) : (A^k)_{ij} > 0\}$ .*

Acting on features  $H \in \mathbb{R}^{n \times d}$ ,  $W^{(k)}H$  sends to each node  $j$  a sum of the features of every node reaching  $j$  by a length- $k$  walk, weighted by multiplicity. The normalisation is global, not by row: row normalisation would flatten the multiplicities to a uniform distribution, erasing what distinguishes  $W^{(k)}$  from a reachability mask. Since  $\mathcal{S}_k$  is the nonzero pattern of  $A^k$ ,  $W^{(k)}$  is stored and applied in  $O(|\mathcal{S}_k|)$ . The fixed-weight  $W^{(k)}$  already mitigates over-squashing—node  $j$  reads range- $k$  sources *directly*, in one step—but its weights cannot *select* which reachable source matters, which motivates Walk Attention.

### 3.5 The quotient algebra

Representation theory routinely passes from  $\mathbb{k}Q$  to a quotient  $\mathbb{k}Q/I$  by an *admissible ideal*  $I$ , identifying paths declared equal. In the diamond of Example 2,  $\beta_1\alpha_1 \equiv \beta_2\alpha_2$  generates an ideal with  $\dim(e_4(\mathbb{k}Q/I)_2e_1) = 1$ ; in general  $\dim(e_j(\mathbb{k}Q/I)_ke_i) \leq n_k(i, j)$ . Using the quotient as a remedy—replacing

multiplicities by the smaller quotient dimensions to “de-duplicate” paths before aggregation—turns out to be *counterproductive* (Section 5): it destroys multiplicity that the task may need. The constructive role of the quotient is to *reshape the support*, e.g. routing path classes to distinct heads (future work); the model below uses only  $\mathcal{S}_k$  and the unquotiented  $W^{(k)}$ .

### 3.6 A common algebraic ground

The graded algebra  $\mathbb{k}Q$ , the impact degree with its FNS, and the quotient  $\mathbb{k}Q/I$  form one substrate read in two ways, according to whether a node-update map  $\varphi$  over the layers  $\mathcal{N}_k(c)$  is prescribed or learned. In the *dynamical* reading  $\varphi$  is a fixed evolution rule weighted by impact degree, and iterating it yields the *automata of impact over quivers* (AIQ) developed separately [25,13]. In the *representational* reading, taken here, a vector space per vertex and a linear map per arrow is a representation of  $Q$  (a  $\mathbb{k}Q$ -module,  $\text{Rep}(Q) \cong \mathbb{k}Q\text{-Mod}$ ), a graph neural network on  $Q$  is such a representation with learned arrow maps, and Walk Attention is its graded instance.

## 4 Walk Attention

We now define the architecture, and we begin with the observation that motivates it: *the walk operator already is an attention*. An attention layer, in its general form, fixes a *support* relation  $S$  of (target, source) pairs and computes, for each target  $j$ , a normalised weighted combination of transformed source features,

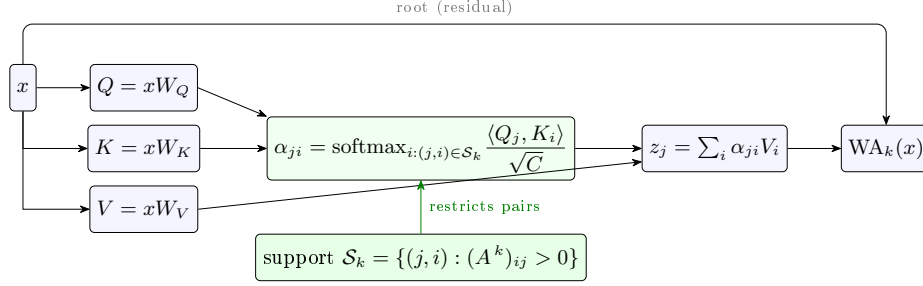
$$\text{out}_j = \sum_{i: (j,i) \in S} \alpha_{ji} Vx_i, \quad \alpha_{ji} \geq 0, \quad \sum_i \alpha_{ji} \text{ bounded}, \quad (1)$$

and architectures differ only in two choices: *which pairs*  $S$  contains, and *where the coefficients*  $\alpha$  come from. Transformer self-attention [20] takes  $S =$  all pairs and learns  $\alpha$  from query–key products; GAT [21] takes  $S =$  the edges and learns  $\alpha$ . Now compare with the walk operator of Definition 5: acting on features it computes  $(W^{(k)}H)_j = \sum_{i: (j,i) \in \mathcal{S}_k} \frac{n_k(i,j)}{Z_k} H_i$ —*exactly the form* (1), with support the walk-reachability  $\mathcal{S}_k$  and coefficients fixed at the normalised walk multiplicities. Aggregating over length- $k$  walks therefore *is* an attention whose support the path algebra determines exactly (Proposition 1) and whose weights are frozen. Walk Attention keeps the algebraic support and replaces the frozen multiplicities by learned query–key weights—the learned node update  $\varphi$  of Section 3.6.

### 4.1 The Walk Attention layer

Fix a range  $k$  with support  $\mathcal{S}_k$ . A Walk Attention layer with  $H$  heads of dimension  $C$  computes queries, keys and values by linear maps,  $Q^h = xW_Q^h$ ,  $K^h = xW_K^h$ ,  $V^h = xW_V^h$ , and attends *only over the reachable pairs*  $(j,i) \in \mathcal{S}_k$ :

$$\ell_{ji}^h = \frac{1}{\sqrt{C}} \langle Q_j^h, K_i^h \rangle, \quad \alpha_{ji}^h = \text{softmax}_{i: (j,i) \in \mathcal{S}_k} \ell_{ji}^h, \quad z_j^h = \sum_{i: (j,i) \in \mathcal{S}_k} \alpha_{ji}^h V_i^h. \quad (2)$$



**Fig. 2.** One Walk Attention layer at range  $k$ . Queries, keys and values are linear maps of the node features; attention is computed *only* over the walk-reachable pairs  $\mathcal{S}_k$  supplied by the path algebra (the nonzero pattern of  $A^k$ ), then aggregated and added to a residual “root” term. The green support box is what distinguishes Walk Attention from dense ( $\mathcal{S}_k = \text{all pairs}$ ) and one-hop ( $\mathcal{S}_k = \text{edges}$ ) attention.

The softmax is taken, for each target  $j$ , over exactly the sources that reach  $j$  by a length- $k$  walk—a *segmented* softmax over the support, never over all  $n$  nodes. The head outputs are concatenated and a residual “root” term is added,  $\text{WA}_k(x)_j = \|_h z_j^h + x_j W_{\text{root}}$ . Because (2) only ever touches pairs in  $\mathcal{S}_k$ , the layer costs  $O(|\mathcal{S}_k|HC)$  time and memory, not  $O(n^2)$ ; the supports are precomputed once per graph as the nonzero patterns of  $A^k$ . The precomputation is itself sparse: the patterns are obtained by  $k - 1$  sparse matrix–matrix products, at a cost governed by the nonzeros of the intermediate powers rather than by  $n^2$ ; on massive graphs whose higher powers densify, the support can be truncated or sampled before attention is applied (Section 6). Figure 2 shows the data flow and Algorithm 1 summarises one layer.

*Example 3.* In the diamond of Example 2 at range  $k = 2$  the support is the single pair  $\mathcal{S}_2 = \{(4, 1)\}$ : node 4 attends directly to node 1, the softmax over a singleton gives  $\alpha_{41} = 1$ , and  $\text{WA}_2(x)_4 = Vx_1 + x_4 W_{\text{root}}$ —the two parallel walks become one *direct* interaction. With a second source  $1'$  (arrows to 2, 3),  $\mathcal{S}_2 = \{(4, 1), (4, 1')\}$  and the softmax weighs the two range-2 sources by key–query match, so node 4 reads both *directly and chooses*; a one-hop network must push both signals through the shared vertices 2, 3—the very compression that over-squashing names.

A depth- $L$  Walk Attention network stacks one layer per range  $k = 1, \dots, L$ , so successive layers read from progressively longer walks; each layer is followed by layer normalisation, an ELU nonlinearity and dropout, and a final MLP maps the embedding to task outputs. Layer normalisation is important in practice: without it the learned attention is high-variance across initialisations; with it, together with a short learning-rate warmup and gradient clipping, training converges reliably (Section 5).

Table 1 situates Walk Attention along the two axes of (1): GAT learns weights but propagates long-range signal through  $L$  stacked layers; a graph Transformer



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**Algorithm 1** Walk Attention layer at range  $k$  (sparse).

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**Require:** features  $x \in \mathbb{R}^{n \times d_{in}}$ ; support  $\mathcal{S}_k = \{(j, i) : (A^k)_{ij} > 0\}$

- 1:  $Q, K, V \leftarrow xW_Q, xW_K, xW_V$   $\triangleright$  per head; shapes  $n \times HC$
- 2: **for** each  $(j, i) \in \mathcal{S}_k$  and head  $h$  **do**
- 3:    $\ell_{ji}^h \leftarrow \langle Q_j^h, K_i^h \rangle / \sqrt{C}$
- 4: **end for**
- 5:  $\alpha_{ji}^h \leftarrow \text{segmented-softmax over } \{i : (j, i) \in \mathcal{S}_k\}$
- 6:  $z_j^h \leftarrow \sum_i \alpha_{ji}^h V_i^h$   $\triangleright$  scatter-add into targets
- 7: **return**  $\|_h z^h + xW_{\text{root}}$

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**Table 1.** Walk Attention against related aggregation operators, by the two axes of (1): support and weights.

| Operator                     | Support               | Weights              | Cost                 |
|------------------------------|-----------------------|----------------------|----------------------|
| GCN / message passing        | 1-hop                 | fixed (degree)       | $O( Q_1 )$           |
| Graph attention (GAT)        | 1-hop                 | learned              | $O( Q_1 )$           |
| Walk operator $W^{(k)}$      | walk-reachable        | fixed (multiplicity) | $O( \mathcal{S}_k )$ |
| Graph Transformer            | all pairs             | learned              | $O(n^2)$             |
| <b>Walk Attention (ours)</b> | <b>walk-reachable</b> | <b>learned</b>       | $O( \mathcal{S}_k )$ |

learns weights over all pairs at  $O(n^2)$ ; the walk operator has the multi-hop support but cannot select. Walk Attention occupies the remaining cell.

## 4.2 A formal account on the bottleneck family

On the bottleneck-chain family used in Section 5, the separation between one-hop message passing and Walk Attention can be *proved*, not only measured. The construction:  $K$  sources carry a one-hot *key*—the key assignment is a uniformly random permutation  $\pi$ —and a one-hot *value* (a distinct basis vector for each source);  $d$  bottleneck layers  $L_1, \dots, L_d$  of  $M$  interchangeable, featureless vertices follow, consecutive layers fully connected; a target carries a one-hot *query*  $q$  and must output the value of the source whose key is  $q$  (label  $\pi^{-1}(q)$ , chance  $1/K$ ). Networks have the standard depth  $L = d + 1$ —one layer per hop, as in [3] and in our experiments—and hidden width  $w$ .

**Proposition 3.** *For any message-passing network whose update  $h'_v = U(h_v, \text{AGG}\{h_u : u \rightarrow v\})$  is shared across vertices: after every round, all  $M$  vertices of each bottleneck layer carry identical states. Hence the entire influence of the sources on the target passes through a single  $w$ -dimensional state per layer—while  $KM^d$  walks carry it (Proposition 1).*

*Proof.* Induction on the round: vertices of a layer start equal (zero features) and share their in-multiset (all sources for  $L_1$ , all of  $L_{r-1}$  for  $L_r$ ), so shared  $U, \text{AGG}$  keep them equal.  $\square$

**Proposition 4.** *If moreover messages are linear in the sender’s state with content-independent coefficients—as in GCN and GIN—then with  $L = d + 1$  layers the target’s output depends on the sources only through  $\sum_s x_s$ , which is the same for every  $\pi$ : the one-hot keys of a permutation sum to the all-ones vector. Expected accuracy is exactly  $1/K$  for every  $d$ ,  $w$ , and parameters.*

*Proof.* Source information entering  $L_1$  at round  $t$  needs  $d$  more hops, reaching the target at round  $t + d \leq d + 1$  only if  $t \leq 1$ ; at round 1 the message from source  $s$  is a fixed linear map of the raw  $x_s$  with one shared coefficient (sources are isomorphic in the graph), so  $L_1$ ’s state is a function of  $\sum_s x_s$ , whose key and value blocks are  $\pi$ -independent. Given  $q$ , the label  $\pi^{-1}(q)$  is uniform, so a  $\pi$ -independent prediction scores  $1/K$ .  $\square$

**Proposition 5.** *At range  $k = d + 1$  the target’s support is exactly the  $K$  sources (the graph is graded; Proposition 2). With  $W_Q$  projecting the query slot scaled by any  $\beta > 0$ ,  $W_K$  the key slot,  $W_V$  the value slot and  $W_{\text{rot}} = 0$ , the output  $z_t = \sum_s \alpha_s e_{v(s)}$  of a single Walk Attention layer has its largest entry at the correct value, on every instance, for all  $d$  and  $M$ .*

*Proof.* The logit of source  $s$  is  $(\beta/\sqrt{C}) \mathbf{1}[\pi(s) = q]$ , so the softmax puts its largest weight  $e^{\beta/\sqrt{C}}/(e^{\beta/\sqrt{C}} + K - 1)$  on the unique match  $s^*$ ; the values being distinct basis vectors, the largest coordinate of  $z_t$  is  $\alpha_{s^*}$ , at  $v(s^*) = \pi^{-1}(q)$ .  $\square$

*Example 4.* Take  $K = 2$ ,  $M = 2$ ,  $d = 1$  and depth  $L = 2$ . The two key assignments,  $\pi = \text{id}$  and the swap, produce the *same* source sum  $\sum_s x_s$  (key and value blocks both  $(1, 1)$ ), so by Proposition 4 a GCN or GIN computes identical outputs on the two instances—whose labels differ: accuracy  $1/2$ , chance. A Walk Attention layer at range 2 attends over  $\{s_1, s_2\}$  with logits  $\beta \langle e_q, e_{\pi(s)} \rangle$ , which distinguish the instances at the first multiplication and select the matching source (Proposition 5).

*Remark 1.* The propositions delimit the mechanisms seen in Section 5. GAT escapes Proposition 4—its first-round coefficients depend on source content—but must still thread the key–value association through the single-vector channel of Proposition 3, re-encoded  $d + 1$  times: empirically it sits above chance and degrades with depth. The fixed-weight walk operator reads the sources directly but with *equal* weights (all multiplicities are  $M^d$ ), so it must embed the whole key–value multiset in  $w$  dimensions rather than select one source: it plateaus in between. Walk Attention both reads directly and selects (Proposition 5), and solves the task.

## 5 Experiments

We evaluate on a controlled family in which over-squashing is severe and tunable, then on two external benchmarks. All code, configurations and experiment logs are released [26].

**Table 2.** Target accuracy (mean  $\pm$  std over 5 seeds) on bottleneck-chain retrieval,  $K=5$ ,  $M=4$ ; chance  $1/K = 0.20$ ; the answer lies  $d+1$  hops away. GCN/GIN sit at chance as Proposition 4 mandates (the residual  $+0.01$ – $0.02$  is best-epoch selection on a finite validation split).

| Model                             | depth $d = 2$                       | depth $d = 3$                       |
|-----------------------------------|-------------------------------------|-------------------------------------|
| GCN (one-hop, linear aggregation) | $0.21 \pm 0.01$                     | $0.22 \pm 0.01$                     |
| GIN (one-hop, linear aggregation) | $0.21 \pm 0.01$                     | $0.21 \pm 0.01$                     |
| GAT (one-hop attention)           | $0.45 \pm 0.22$                     | $0.29 \pm 0.17$                     |
| Walk operator (fixed weights)     | $0.62 \pm 0.12$                     | $0.50 \pm 0.12$                     |
| <b>Walk Attention (ours)</b>      | <b><math>1.000 \pm 0.000</math></b> | <b><math>1.000 \pm 0.000</math></b> |

*Bottleneck-chain retrieval.* We instantiate the family of Section 4.2 with  $K=5$ ,  $M=4$  and depths  $d \in \{2, 3\}$ : the walk multiplicity into the target is  $KM^d$  (Proposition 1), while the deepest support consists of only the  $K$  source–target pairs (2.6% and 1.5% of node pairs at  $d=2, 3$ ). All models use hidden width 16, four heads where applicable,  $d+1$  layers/ranges, and AdamW with a ten-epoch warmup, cosine decay and gradient clipping; Walk Attention adds layer normalisation. We report validation target accuracy (no separate test split), mean  $\pm$  std over five seeds. Table 2 lands exactly where Propositions 3–5 put it (Remark 1): GCN and GIN are pinned at chance and sit there empirically; GAT degrades sharply with depth; the walk operator crosses the bottleneck in one step but, weighting all sources by multiplicity alone, cannot select the matching one and plateaus; **Walk Attention solves the task exactly** at both depths over all seeds. Without layer normalisation and warmup its accuracy ranged 0.44–1.00 across seeds; the standard stabilisers remove this variance entirely.

*RingTransfer.* The bottleneck family is our own construction; to confirm the effect on an independent benchmark we use RINGTRANSFER [5,7]: on a cycle of  $n$  nodes a source must transmit its one-hot label (five classes, chance 0.2) to the antipodal target,  $n/2$  hops away along the two arcs. Every node has two neighbours, so the number of length- $(n/2)$  walks entering the target grows as  $2^{n/2}$  while only the two arc walks carry the signal. All models use hidden width 32, the same optimiser recipe and  $n/2$  layers/ranges. Table 3: small rings are easy for every model; from  $n=14$  the one-hop baselines degrade and become seed-dependent—at  $n=18$  GAT falls to  $0.29 \pm 0.19$  and GCN to  $0.53 \pm 0.33$ —while **Walk Attention stays at  $1.000 \pm 0.000$  throughout**: the gap widens with the severity of over-squashing, as predicted.

*Collapsing walks hurts.* The quotient  $\mathbb{K}Q/I$  suggests an alternative remedy: replace  $n_k(i, j)$  by the smaller quotient dimensions, de-duplicating equivalent walks before aggregation. An ablation that changes *only* the fixed-weight operator (Table 4) shows the quotient falling clearly below the raw operator at both depths: collapsing parallel walks discards the source multiplicity the retrieval task needs. Here redundant paths are *signal*; the algebra’s contribution is the support, not

**Table 3.** RINGTRANSFER accuracy vs. ring size  $n$  (distance  $n/2$ ), mean  $\pm$  std over 5 seeds; chance 0.20.

| Model                        | $n=6$                             | $n=10$                            | $n=14$                            | $n=18$                            |
|------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| GCN                          | $1.00 \pm 0.00$                   | $1.00 \pm 0.00$                   | $0.81 \pm 0.13$                   | $0.53 \pm 0.33$                   |
| GAT                          | $1.00 \pm 0.00$                   | $1.00 \pm 0.00$                   | $0.69 \pm 0.29$                   | $0.29 \pm 0.19$                   |
| <b>Walk Attention (ours)</b> | <b><math>1.00 \pm 0.00</math></b> | <b><math>1.00 \pm 0.00</math></b> | <b><math>1.00 \pm 0.00</math></b> | <b><math>1.00 \pm 0.00</math></b> |

**Table 4.** De-duplicating walks via  $\mathbb{k}Q/I$  *hurts*: mean  $\pm$  std over 5 seeds, same protocol as Table 2 (the walk-operator row coincides).

| Operator                                   | depth $d = 2$                     | depth $d = 3$                     |
|--------------------------------------------|-----------------------------------|-----------------------------------|
| Quotient $\mathbb{k}Q/I$ (collapsed walks) | $0.39 \pm 0.04$                   | $0.30 \pm 0.09$                   |
| Walk operator (raw walks)                  | <b><math>0.62 \pm 0.12</math></b> | <b><math>0.50 \pm 0.12</math></b> |

compression. Whether a regime exists where compression helps—genuinely redundant paths, or path classes routed to distinct heads—remains open.

*LRGB Peptides-func.* The synthetic benchmarks are deliberately worst cases; to assess real data we use Peptides-func [8] (10,873/2,331/2,331 molecular graphs, median 138 nodes; multi-label classification, test Average Precision), with the same atom embedding, four layers, hidden width 80, mean pooling and AdamW for every model; Walk Attention uses ranges  $k = 1, \dots, 4$  (support 1–8% of  $n^2$  at range 3). Table 5: Walk Attention exceeds GCN and ties GAT ( $0.526 \pm 0.007$  vs.  $0.526 \pm 0.004$ , four seeds) at about  $1.6\times$  its wall-clock. All models share a deliberately small budget, so absolute numbers sit below published results (GCN 0.593 under the tuned 500k-parameter protocol [8], higher in re-evaluations [18]): the comparison is *internal*, isolating the operator. The tie is not for lack of parallel paths but because over-squashing is *mild* here; the next paragraph makes this quantitative.

*Predicting the regime.* The capacity argument behind Propositions 3–4 yields a measurable quantity. After  $k$  rounds every length- $k$  walk into  $j$ —one basis element of  $e_j \mathbb{k}Q_k = \bigoplus_i e_i \mathbb{k}Q_k e_i$ —must be encoded in a  $w$ -dimensional state, which separates of the order of  $\log_2 w$  bits. We define the *severity index*

$$\Psi_k(j) = \log_2 \dim_{\mathbb{k}}(e_j \mathbb{k}Q_k) - \log_2 w = \log_2 \sum_i n_k(i, j) - \log_2 w, \quad (3)$$

the excess of incoming walk entropy over representation capacity at the decision target and range. It is computed from the graded algebra alone, by the vector iteration  $\mathbf{1}^T A^k$  in  $O(k|Q_1|)$  without forming  $A^k$ ; in closed form  $\Psi = \log_2(KM^d) - \log_2 w$  on the bottleneck family and  $\Psi = n/2 - \log_2 w$  on the ring. Table 6 sets  $\Psi$  against the measured advantage on every configuration above and on four run for this purpose (ring  $n \in \{12, 16, 20\}$ , bottleneck  $d=4$ , same protocol). The relation is a threshold rather than a trend: every configuration

**Table 5.** LRGB Peptides-func, test AP (mean  $\pm$  std over 4 seeds), shared small budget isolating the operator; published results under the standard 500k protocol are higher for every architecture [8,18].

| Model                 | test AP           |
|-----------------------|-------------------|
| GCN                   | $0.492 \pm 0.011$ |
| GAT                   | $0.526 \pm 0.004$ |
| Walk Attention (ours) | $0.526 \pm 0.007$ |

**Table 6.** Severity index  $\Psi$  (bits) against the advantage of Walk Attention over the best one-hop baseline (mean over 5 seeds). Rn: RINGTRANSFER of size  $n$  ( $w=32$ ); Bd: bottleneck of depth  $d$  ( $w=16$ ); Pep: Peptides-func ( $w=80$ ). Spearman  $\rho = 0.875$ .

|           | R6  | Pep   | R10 | R12 | R14  | B2   | R16  | R18  | B3   | R20  | B4   |
|-----------|-----|-------|-----|-----|------|------|------|------|------|------|------|
| $\Psi$    | -2  | -0.75 | 0   | 1   | 2    | 2.3  | 3    | 4    | 4.3  | 5    | 6.3  |
| advantage | .00 | .00   | .00 | .00 | +.19 | +.55 | +.47 | +.47 | +.71 | +.48 | +.63 |

with  $\Psi \leq 1$ —including Peptides-func at  $\Psi = -0.75$ , where a four-range support at width 80 is not saturated—has advantage exactly zero, and the advantage appears at  $\Psi = 2$  and stays large beyond (Spearman  $\rho = 0.875$ ,  $p = 4 \times 10^{-4}$ ,  $n = 11$ ). Near the threshold the one-hop baselines are bimodal across seeds (at  $n=16$ , GCN scores  $\{0.61, 0.21, 0.21, 0.81, 0.82\}$ ): they either encode the signal or collapse to chance, whereas Walk Attention is exact on every seed. The index is built from multiplicities, which the binary support of  $A^k$  does not see.

*A real network with severe squashing.* Used prospectively,  $\Psi$  selects real data. On the cit-HepPh citation quiver [14] (34,546 papers) the incoming walk entropy reaches 21 bits at range 4, with pairs joined by 22,702 parallel walks, so  $\Psi \approx +15$  at width 64. We build an early-impact benchmark on it: the citation graph up to 1997 (13,745 papers, 98,307 arrows), node features restricted to publication year and in/out-degree at the cut, and five binary targets on the 6,635 papers of 1996–97 (future citations within degree stratum, residual growth beyond degree and age, independence of future citers, persistence, growth of the impact neighbourhood  $A_2(c)$ ), scored by balanced accuracy over five stratified splits. With the support truncated to the 32 sources of largest multiplicity per target and range ( $\text{nnz} \leq 32n$  per range), Walk Attention is the best model on four of the five targets, by a small but consistent margin over GAT with residual connections (+0.009 balanced accuracy on average, 15/20 paired target–seed comparisons, Wilcoxon  $p = 0.004$ ; +0.016 over GCN,  $p < 10^{-3}$ ). The margin is far smaller than on the synthetic families although  $\Psi$  is far larger, which sharpens the reading of the index:  $\Psi$  measures how much walk information the local channel loses, not how much of it the label requires. It is a necessary condition for an advantage—below the threshold none is possible—while the magnitude is set by the task’s dependence on deep structure: total in the retrieval families,

modest in impact prediction, where degree already summarises most of what is predictable.

## 6 Conclusion and Future Work

We introduced Walk Attention, an architecture that mitigates over-squashing by attending over a multi-hop support supplied by the path algebra of the underlying quiver. Proposition 1 gives the support and its multiplicities exactly; Propositions 3–5 prove the separation on the bottleneck family—message passing with linear first aggregation is at chance for every depth and width, while one Walk Attention layer at the matching range is exact—and the experiments confirm it. The severity index  $\Psi$ , computed from the graded dimensions alone, turns the regime dependence observed on RINGTRANSFER and Peptides-func into a prediction (no advantage below 1–2 bits of excess entropy, a large one above) and selects a real citation network on which Walk Attention is the best of the compared models by a small, consistent margin. A complementary negative result showed that compressing equivalent walks via the quotient is counterproductive when multiplicity carries signal: the role of the algebra is to determine the attention support, not to discard information. The construction is deliberately elementary and self-contained, an entry point connecting quiver representation theory to graph representation learning; it is the learned reading of a common substrate whose dynamical reading is the AIQ framework [25]. Several directions follow.

- **From learned weights to dynamics.** Inject the attention weights learned on a task into an AIQ as a data-driven evolution rule, reading off the dynamical effect of each weight, and conversely let AIQ stability and reachability analyses interpret and constrain what Walk Attention has learned—to us the most promising bridge.
- **Benchmarks with severe over-squashing.** Real long-range tasks beyond citation networks, selected by  $\Psi$ , compared against rewiring [19], multi-hop [12] and graph-Transformer baselines.
- **Quotient-shaped supports.** Use  $\mathbb{k}Q/I$  to *partition* walks into equivalence classes routed to distinct attention heads, rather than to compress; identifying which relations  $I$  are useful, and learning them from data, is an appealing problem. A concrete instance: in a heterogeneous graph whose walks compose typed relations, an ideal identifying the semantically equivalent compositions would tie each class to one shared head—sharing weights across redundant paths while keeping genuinely distinct influences apart.
- **Scaling the support.** Truncate, sample, or learn sparser admissible supports on dense graphs, where the range- $k$  support can itself grow large.

**Reproducibility.** All graph generators, the walk operators, the Walk Attention layer, the baselines and the experiment logs are released [26]; the path-algebra computations build on an open-source path-algebra library [24].

**Use of AI Tools.** The authors used Claude (Anthropic) as an assistant for code, experiment execution, drafting and editing; all scientific decisions, analyses and claims were made and verified by the authors, who take full responsibility for the content.

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